

CS6440: Introduction to Health Informatics Spr24

Practicum Sprint #4 - Status Check In #2

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Abstract—We introduce *SkinWise*, a web-based diagnostic tool enhancing healthcare providers' capabilities in early skin cancer detection and management, focusing on melanoma. Leveraging advanced machine learning techniques and datasets like Skin Cancer ISIC and FHIR, *SkinWise* offers innovative solutions for accurate cancer classification and management.

1 ACCOMPLISHMENTS

In alignment with the tasks outlined in our Sprint #3, we kept working on the planned tasks within the specified timeframe. [Table 1](#) is a summary of our achievements, indicating the primary contributors within our team. As a team, we tackled hurdles collectively, fostering a culture of teamwork. Moreover, to ensure cohesion and progress, we have regular group meetings for internal deliberations and strategic planning and use the [GitHub](#) repository to maintain updates and version control. Overall, we aim to create a streamlined process and guarantee smooth teamwork across team members spanning various time zones.

Project Tasks / Activities	Group Member
Refine the codebase for our web app	Jun, Zhiqiu, Zifeng
Web app deployment setup (Locally and also on AWS)	Jun, Zhiqiu, Zifeng
Explore how the FHIR skin patient data can be connected with the classification model	Zifeng
Examine the performance of a selected image classification model	Jun, Zhiqiu
Debug the codebase for our web app	Jun
Test local deployment for the CRUD interface for Patient Data Management	Jun, Zhiqiu, Zifeng

Table 1—A list of accomplishments

2 CHALLENGES

In the following, we describe the primary challenges encountered thus far during this sprint.

- **Connect the FHIR database with our web application:** One of our primary

challenges lies in seamlessly integrating the FHIR database with our web application. This entails establishing a connection between the two systems to ensure smooth data exchange and functionality. To tackle this challenge, we've explored a potential solution available at [FHIR Streamlit Template](#). This template potentially offers a streamlined approach to incorporating FHIR capabilities into our web application. We will explore if it could help to establish efficient database connectivity and interaction.

- **Local deployment setup using Docker image:** Although we successfully configured a Docker image for local deployment (see [Figure 1](#)), our application remains a work in progress. Despite the achievement of setting up a Docker environment, challenges persist in finalizing the functionalities of our web application. As we continue development, ensuring seamless integration of new features and maintaining compatibility within the Docker environment poses ongoing challenges.
- **Test the efficacy of the selected classification model on skin data:** Evaluating the effectiveness of our chosen classification model on skin data poses a challenge. We found there are some publicly available pre-trained models published on Kaggle, but it requires rigorously assessing how well the model performs in accurately classifying skin-related features, considering factors such as data complexity and model adaptability. So far, we selected the [Melanoma Classification ResNet](#) with a prediction accuracy around 90%, but we have yet to finalize the choice of classification model for our web application.
- **Aligning data input formats between the exported FHIR data and the classification model:** We found that we need to align the data exported from FHIR and the disease classification model. The classification model requires specific data input formats, such as .JPEG format which is required in the tentative selected [Melanoma Classification ResNet Model](#), while the data obtained from FHIR is in .JSON, which needs to be converted into these corresponding formats.
- **Bugs:** We need to fix bugs in the codebase to ensure it works for different data as a comprehensive web-based diagnostic tool as planned.

The above challenges encompass technical obstacles encountered during Skin-Wise's deployment, we are addressing them in the upcoming sprint to make further progress.

3 SPRINT PLANS

Our next check-in is scheduled for two weeks from now. With our progress closely aligning with our projected timeline (see Table 2), we remain committed to adhering to our implementation plan for the tasks outlined in the next sprint period.

4 DEPLOYMENT

By Sprint 4, our focus shifts towards setting up the deployment of our application. At this stage, we tried to identify some suitable deployment solutions for our project. Although our application has not been fully completed, we successfully setup the local deployment for showcasing our progress and facilitating feedback (see Figure 1).

We also successfully deployed our Streamlit application on an Amazon Web Services (AWS) Elastic Compute Cloud (EC2) instance utilizing Docker for containerization (see Figure 2). This process involved setting up an EC2 instance with Amazon Linux 2, configuring the security group to allow inbound traffic on specific ports, and installing Docker. We then cloned the application from a Git repository hosted at <https://github.gatech.edu/jhuang709/SkinWise.git>, built a Docker image from your Dockerfile, and ran the container to serve the Streamlit app. Finally, by mapping port 8501 inside the container to the same port on the EC2 host, we made the application accessible via the instance's public IP and DNS (<http://13.60.53.97:8501> and <http://ec2-13-60-53-97.eu-north-1.compute.amazonaws.com:8501>), showcasing a practical application of cloud computing and containerization for deploying Python applications.

In the next Sprint, we will continue working to implement the necessary steps for deployment and fix bugs in the application.

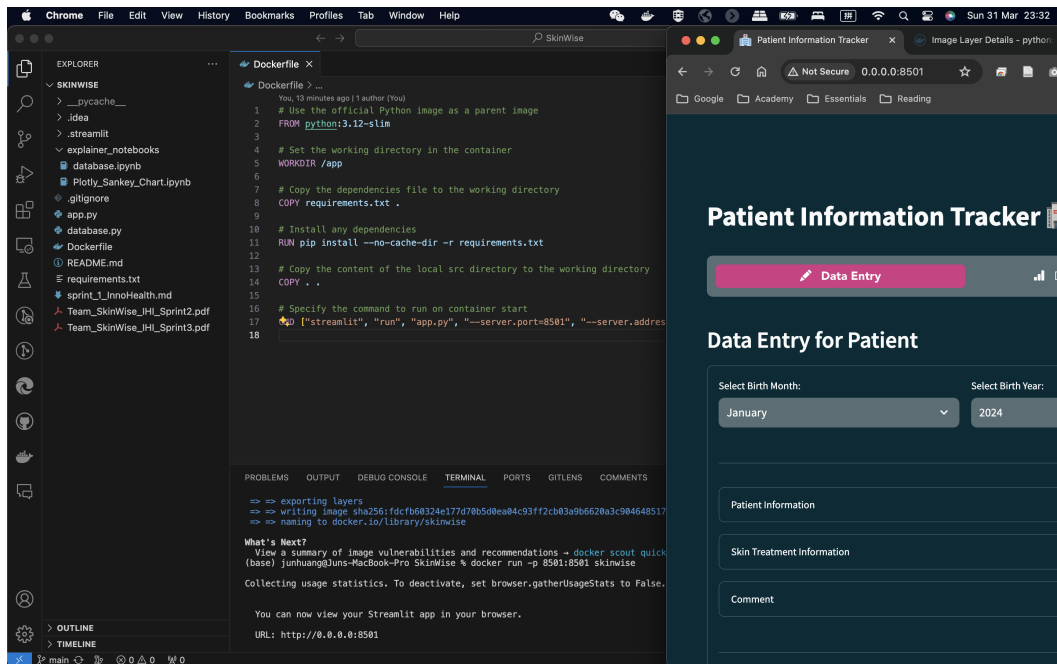


Figure 1—Screenshot of the Docker local deployment of our web app

Patient Information Tracker

 Data Entry

 Data Visualization

Data Entry for Patient

Select Birth Month:

January

Select Birth Year:

2024

Patient Information

Name:

Age:

0

Diagnosis:

Skin Treatment Information

Treatment Plan:

Medication:

Follow-up Schedule:

Comment

Enter a comment here ...

Save Data

Figure 2—Screenshot of the app interface hosted on AWS via publicly accessible link

Project Timeline	Project Tasks	Needs	Risks
Jan 8 - Jan 21	Work to form our team.	Identify potential team members with similar project interests and diverse skills and expertise. Establish clear communication channels for team collaboration.	Difficulty in finding suitable team members with the close interests. Communication challenges within the team, e.g., time zones.
Jan 22 - Jan 28	Team formation and project kick off. Submit Practicum sprint # 1.	Formalize team structure and roles. Define project goals and deliverables.	Lack of clarity in project objectives leading to confusion among team members. Initial setbacks in team coordination and workflow.
Jan 29 - Feb 18	Brainstrom team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.	Research resources and literature relevant to our SkinWise project. Brainstorm the solution. Assign specific roles and responsibilities to team members.	Difficulty in finding relevant and reliable literature/resources. Difficulty in finding relevant health data sources. Potential conflicts arising from role assignments within the team.
Feb 19 - Feb 25	Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.	A project plan outlining the scope, objectives, and timeline of the SkinWise web application. Discussions and consultations with mentor to gather feedback. Explore various architecture design options for the SkinWise. Update our sprint # 2 with mentor's constructive feedback.	Difficulty in finding necessary data sources. Challenges in having interoperability. Shifting project focus leading to scope creep.
Feb 26 - Mar 3	Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.	Detailed project planning and design. Initiate data processing from the Skin Cancer ISIC Dataset. Set up the web app project environment.	Delays in project planning leading to a rushed implementation phase. Technical challenges in processing and handling the dataset. Technical challenges in setting the environment.
Mar 3 - Mar 10	Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.	Development of image classification model using deep neural networks. Initial setup of the web app codebase.	Complexity in implementing deep neural networks for image classification. Technical challenges in the web development with connecting with different datasets.
Mar 11 - Mar 17	Call for the patient resources from FHIR. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.	Access patient resources from FHIR for integration. Refine web app codebase. Conduct mentor check-in to ensure project alignment.	Delays or issues in obtaining necessary patient resources. Integrating data from FHIR sources can be complex. Relying on FHIR resources introduces dependencies.
Mar 18 - Mar 24	Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.	Continued project development including integration of FHIR data. Progress in web app development.	Difficulties in connecting FHIR patient data with the classification model. Challenges in the development of the web app functionality.
Mar 25 - Mar 31	Testing and validation of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.	Testing and validation of the web app. Final mentor check-in before project submission.	Inadequate testing leading to undiscovered bugs or issues. Insufficient time for addressing feedback from mentor check-in.
Apr 1 - Apr 7	Deployment of the SkinWise web application with CRUD functionality	Deployment of the SkinWise web application using CRUD operations. Examine the deployment online.	Deployment issues causing downtime or functionality failures. issues with data integrity, such as data loss or corruption, especially during data migration or database updates. Incomplete deployment leading to user dissatisfaction.
Apr 8 - Apr 14	Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.	Finalize project details and address any remaining issues. Prepare for project submission.	Online deployment may result in performance degradation, such as slow response times, increased latency, or system crashes, especially under heavy loads. Other challenges in the deployment such as configuration errors, network issues.
Apr 15 - Apr 21	Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.	Submission of the final project deliverables.	Technical difficulties or errors during submission. Incomplete or insufficient project documentations.

Table 2—Implementation plan: project tasks, project timeline, and identified needs/risks